

Jigsaw: Graphical Representation for Big Data Management in Digital Film Production

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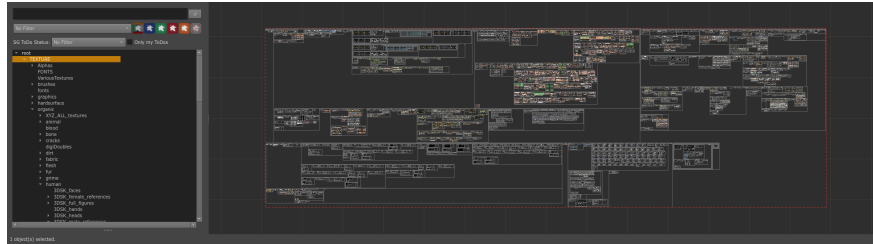


Figure 1: Jigsaw is capable of loading thousands of images in a few seconds and present them in a structured fashion

ABSTRACT

Modern digital film productions rely on huge amounts of supplementary asset data to coordinate the composition of filmed and computer generated content. We present the latest evolution of Jigsaw - DNEG's flagship software used to organise, process, and distribute a wide range of asset data throughout our pipeline. Jigsaw features an intuitive graphical representation of large data-sets, automatic data classification, easy batch processing, seamless pipeline publishing, smooth interactivity, API extendability, and global multi-user coordination. To achieve this, Jigsaw was designed with optimised resource allocation between local and server processing. Large image data-sets are cached on the server as multiresolution JPEGs to be displayed as small thumbnails if viewed all at once or at higher resolution if viewed in detail. Resource intensive tasks are offloaded to the rendering farm for smooth, continued use. To date, Jigsaw has been used in over 300 DNEG shows to ingest, process, and distribute huge amounts of heterogeneous data. As Jigsaw is interdependent on DNEG's pipeline it is limited to online use, away from the production set where asset data is captured and prepared. Future developments are focused on solving this limitation.

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1 INTRODUCTION

Achieving a visually convincing and spectacular VFX composite requires more information about the scene and its contents than provided by the primary film camera image alone. As such, modern productions rely on huge amounts of supplementary asset data to coordinate the seamless composition of filmed and computer-generated content. This data takes the form of concept art, textures, high dynamic range images (HDRIs), photogrammetry data, 3D coordinates/cameras, lens/light information, reference images/videos, and set drawings, to name a few. The Shoot VFX team is responsible for capturing, preparing, ingesting, processing, distributing, and storing this large, and continuous, stream of heterogeneous data. Furthermore, there is very little room for mistakes as the set, and availability of the actors, constantly changes with the progression of the production. Recognizing this, DNEG has invested in the development of Jigsaw, a data management application that has been used in over 300 shows since its initial deployment [et al. 2015].

2 JIGSAW APPLICATION

Jigsaw is used by the Shoot VFX team to organise, process, and distribute a wide range of asset data across various departments. For example, the Lighting department would require HDRIs while Build needs access to 3D photogrammetry/LiDAR assets. To address these requirements the development of Jigsaw prioritised intuitive data navigation of large and heterogeneous data-sets, automatic data categorization, optimised demand of local processing, and global pipeline integration.

2.1 Data navigation

One of the unique features of Jigsaw is its file navigation interface and its ability to handle tens of thousands of files: panning, zooming, and selecting areas of interest as required. Using an intuitive box representation, where folders constitute rectangular groupings of thumbnails, the contents of huge folder structures can be viewed, selected, and organised within a single window, as seen in figure 1.

2.2 Auto Categorization

Data comes in many forms: for example, raw data for HDRI lighting reference might consist of 21 individual photos in a bracketed sequence. A single show can require hundreds of HDRIs, resulting in thousands of images contained within one or more folders. To address this, Jigsaw scans for data patterns, metadata, naming conventions, and machine learning [et al. 2019] to automatically organise content. With a single user action, data can be grouped into their respective HDRI folders, complete with their respective colour charts and slate information. Similarly, texture, prop, and character images, captured by different team members in disparate folders, can be scanned and distributed to their appropriate folders ready for consumption by the respective departments. Heterogeneous data that do not fit a certain category will be ingested as reference data.

2.3 Farm processing

Processing raw data into meaningful assets such as panoramic images and 3D photogrammetry models is most conveniently done through the Jigsaw navigation platform. However, these tasks are complex and resource intensive which may inundate the local machine. To address this, such processes are launched as processing jobs on DNEG's render farm. This maintains local resources free allowing Jigsaw to run smoothly while heavy processes are run in parallel on the backend. Once jobs are done, the Jigsaw data set is updated with the processed assets - immediately available for artists to use.

2.4 Pipeline integration

Asset data is only useful if distributed properly across the organisation. Material such as set information, character/prop scans, and texture images needs to be associated with the respective shots where, and as, it would appear. To achieve this, Jigsaw interacts with Ivy, the DNEG's shots and assets database. This allows the on-set VFX team to publish processed assets into the appropriate show directories ready for artists to use.

3 JIGSAW ARCHITECTURE

Jigsaw's architecture was carefully designed to allow for continuous use by hundreds of employees across all of DNEG. Jigsaw is a Linux based application developed in C++ with a Python API for convenient show-specific customizations. The application uses multiresolution image storage (similar to mipmaps) and a caching and RAM management system for rapid loading and displaying of large datasets.

3.1 Python API for show specific customizations

Jigsaw is extensible using a Python API, giving tremendous flexibility for show-specific customizations. To that end, Jigsaw implements a scripting console and bindings from Qt. The core functionality of this API is managed by the **pybind** library. Jigsaw keeps a list of Python scripts paired with a caller function. When a scripted action is launched from the UI, the caller function is gathered and the Python script is executed.

3.2 Multiresolution cache storage

Work on DNEG shows is typically split across our global sites. This means that hundreds of global users open tens of thousands of images several times a day. A naive server storage implementation would result in terabytes of downloaded data, substantial demand for server resources, and long project startup time. To mitigate this, Jigsaw processes the data on arrival during a nightly job by computing a mipmap image of the dataset, which is then stored in a highly compressed database format on the server. The next user to open the project would only require loading the low resolution portion of the cache into RAM, significantly lowering the demand of local and server resources. Zooming in on areas of interest triggers the loading of higher resolution portions of the cache.

4 RESULTS : USER FEEDBACK

Jigsaw was developed as a direct response to the demands of DNEG's Shoot VFX Department. Since its first development, it has grown in functionality and has tremendously helped the company manage asset data efficiently. Here is what some of our users have to say:

James Porter, Head of Shoot Montreal *"We receive so much data from clients that it can become overwhelming at times. Jigsaw makes things much easier to see what's been delivered, if it's to DNEG-spec and if we've ingested it into the DNEG pipe."*

Kathleen Squire, Shoot Supervisor *"Often so much of what gets done on set gets lost post-shoot, because supervisors and artists just aren't aware it exists. Centralising all that into Jigsaw gives it much better visibility."*

Joe Bannister, Lead Shoot TD *"Many ex-colleagues who have worked at DNEG for Shoot and even other departments have told me how much they miss Jigsaw."*

5 FUTURE DEVELOPMENTS

Jigsaw is DNEG's flagship data management tool and it is continuously evolving. However, it is interdependent on our proprietary pipeline and secure intranet. This makes it virtually impossible to use on set where asset data capture and preparation occurs. Our next evolution of the tool is a stand-alone offline application where the on-set VFX team can immediately view their captured data and conveniently prepare it for pipeline ingestion.

ACKNOWLEDGMENTS

Jigsaw would not be the feature rich application that it is without the continuous feedback of its passionate users. We would like to thank Kieron Helsdon for driving Jigsaw's development for more than 6 years; Kathleen Squire, Joe Bannister and especially James Porter for their engagement in Jigsaw's development and their support in writing and reviewing this paper.

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